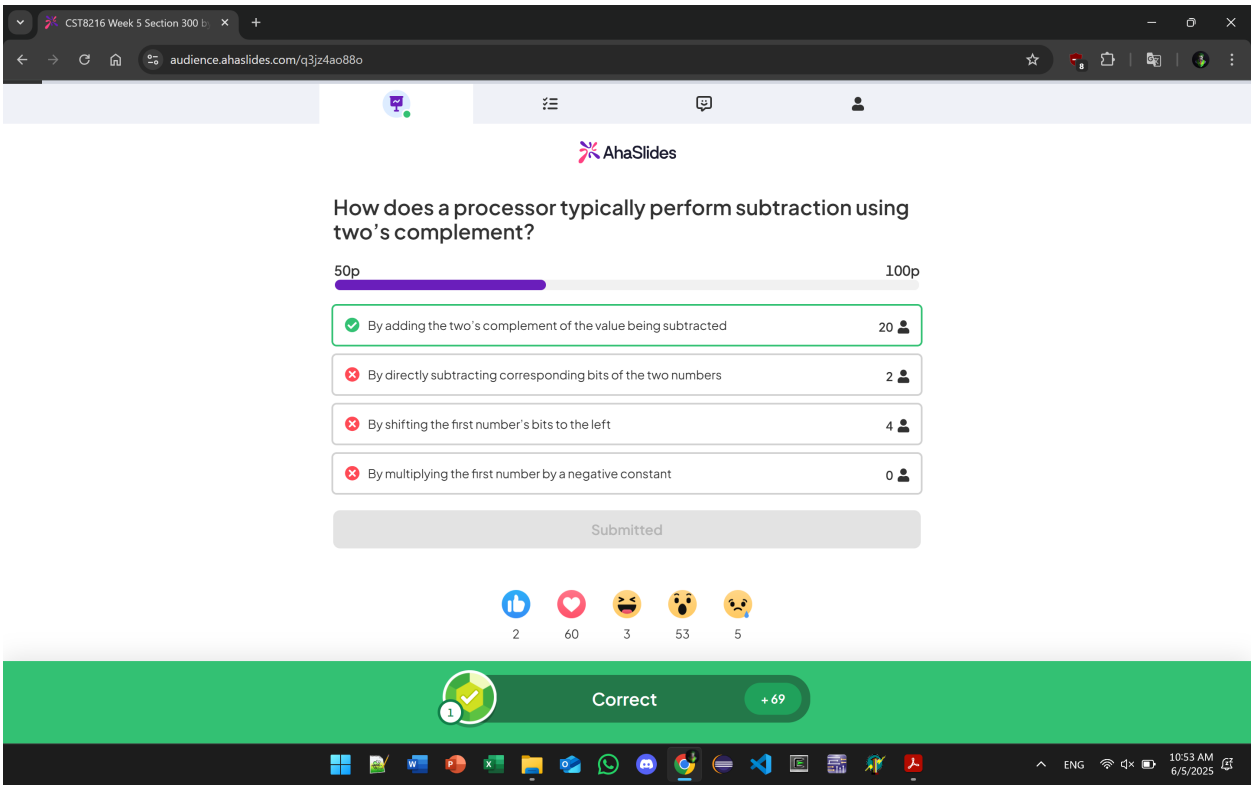
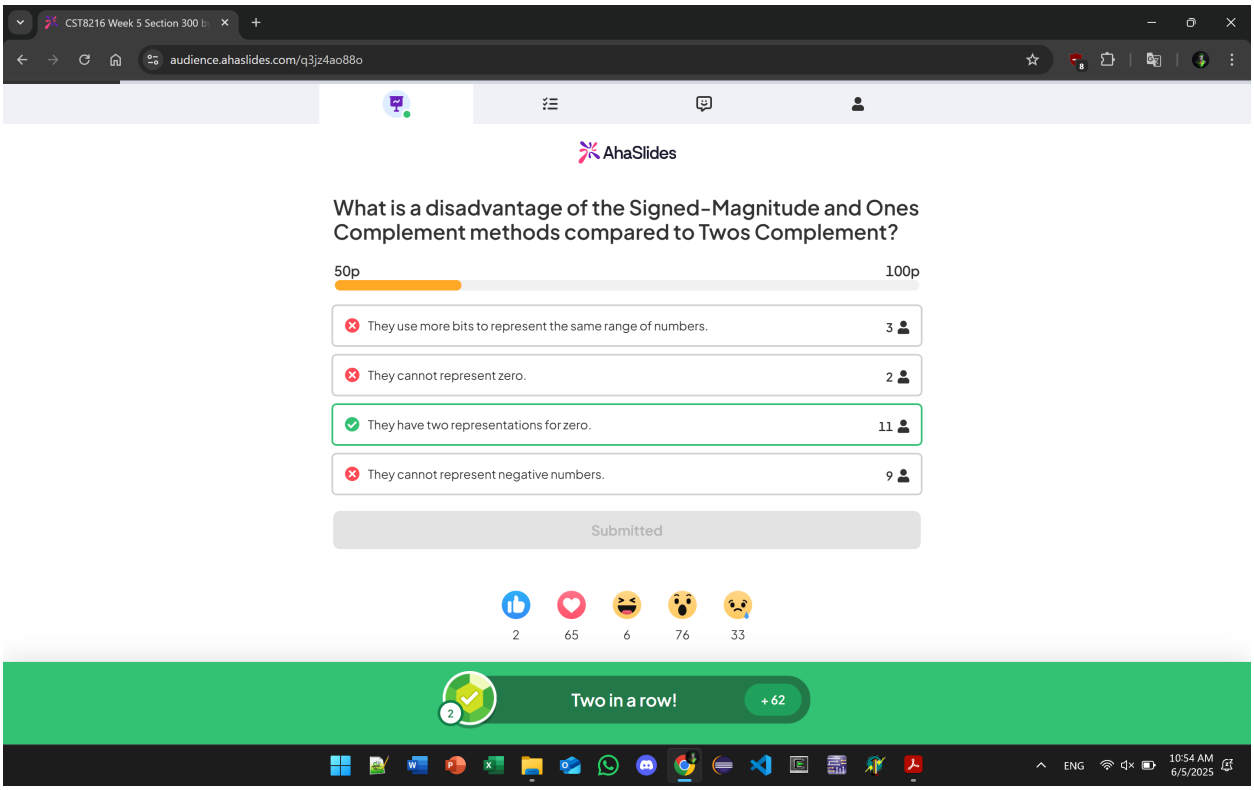
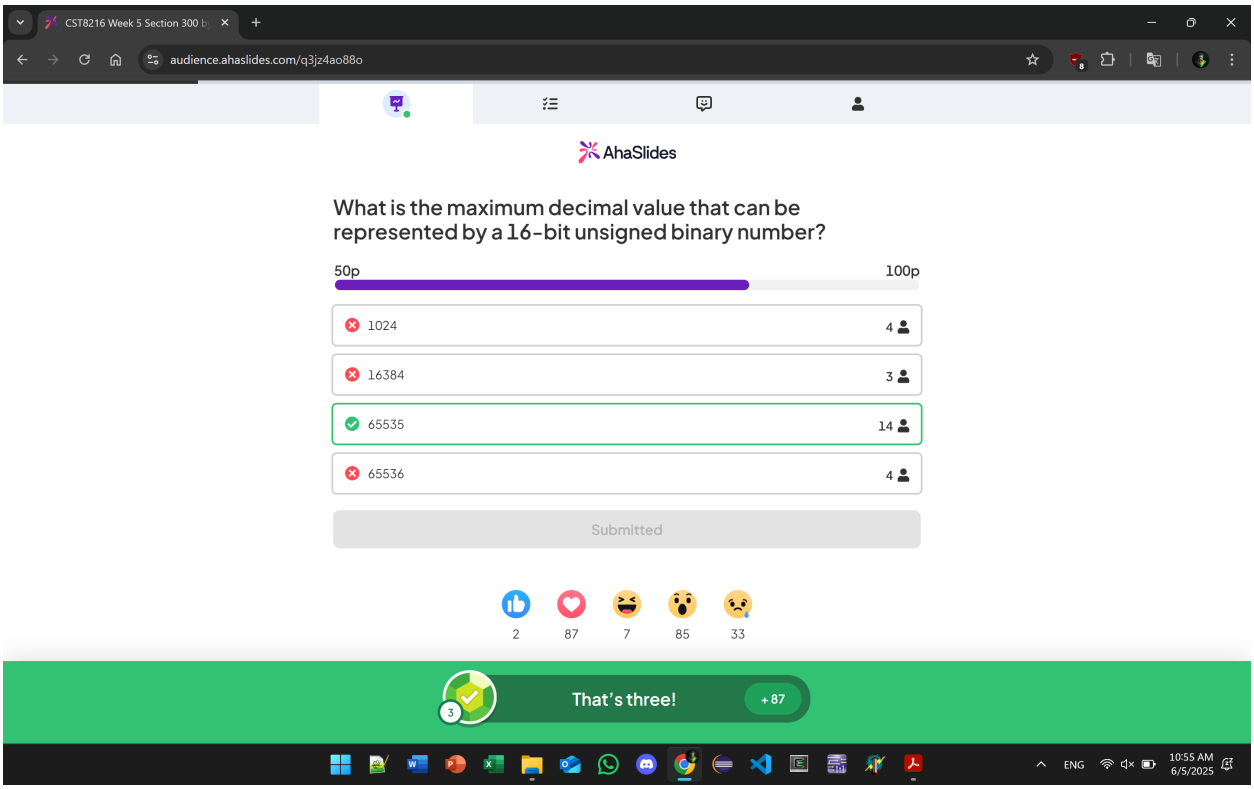
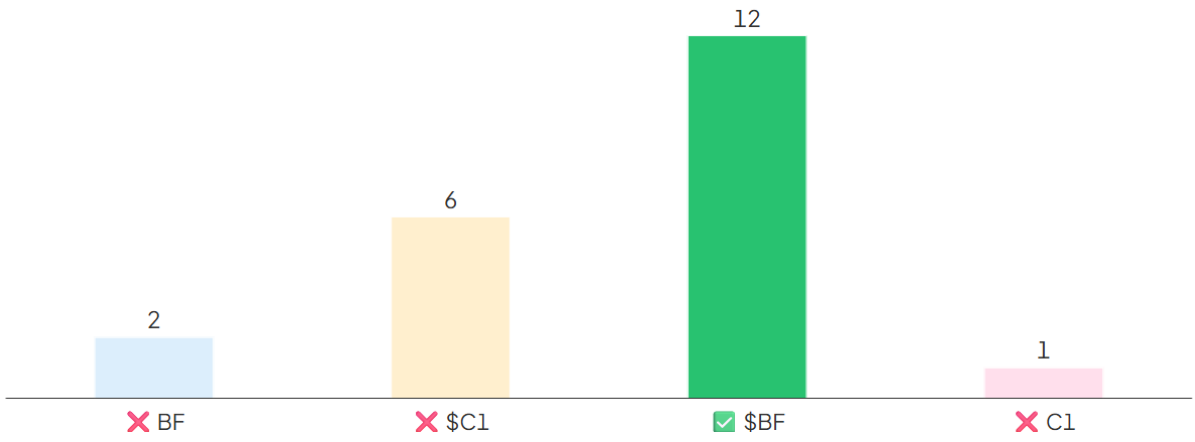

CST8216 Week 5 Section 300 b
audience.ahaslides.com/q3jz4ao88o
AhaSlides
How does a processor typically perform subtraction using two's complement?
50p
100p
By adding the two's complement of the value being subtracted 20
By directly subtracting corresponding bits of the two numbers 2
By shifting the first number's bits to the left 4
By multiplying the first number by a negative constant 0
Submitted
2 60 3 53 5
1
Correct +69
ENG
10:53 AM
6/5/2025

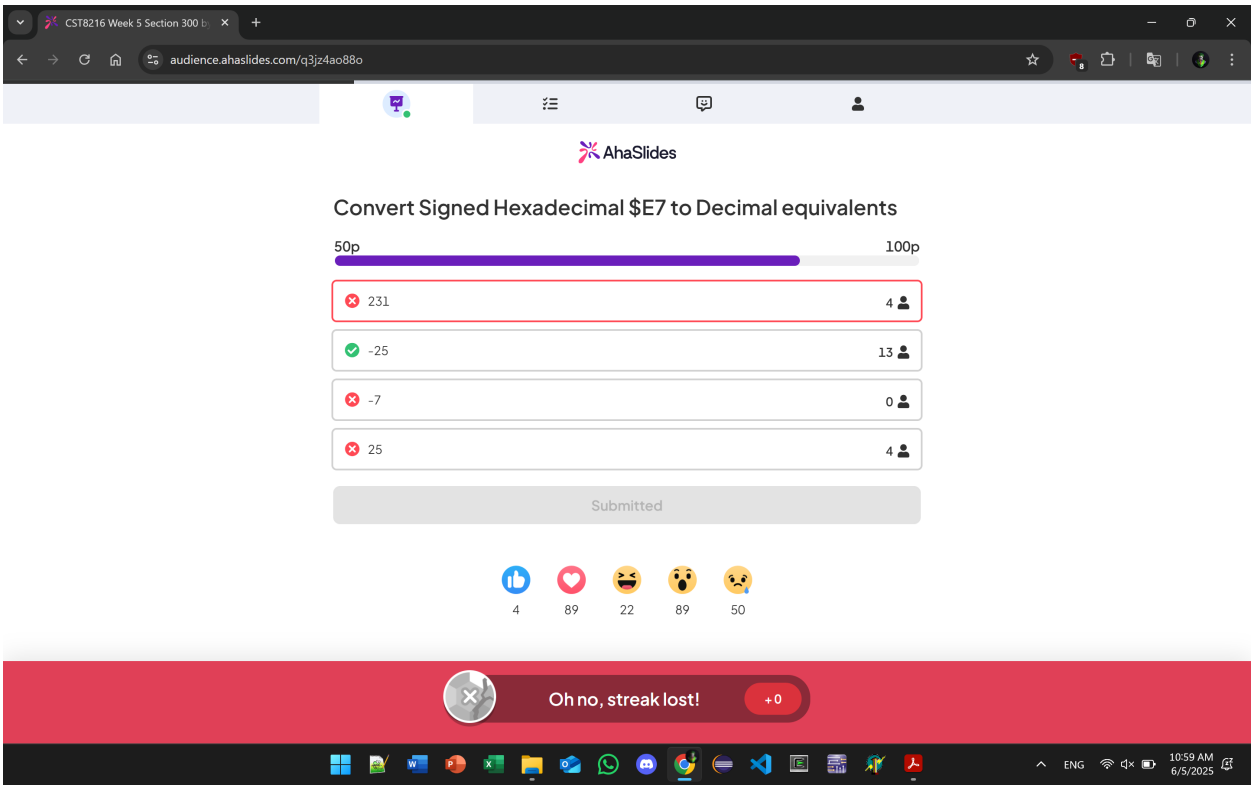

CST8216 Week 5 Section 300 b
audience.ahaslides.com/q3jz4ao88o
AhaSlides
What is a disadvantage of the Signed-Magnitude and Ones Complement methods compared to Twos Complement?
50p
100p
They use more bits to represent the same range of numbers. 3
They cannot represent zero. 2
They have two representations for zero. 11
They cannot represent negative numbers. 9
Submitted
2 65 6 76 33
2
Two in a row! +62
ENG
10:54 AM
6/5/2025

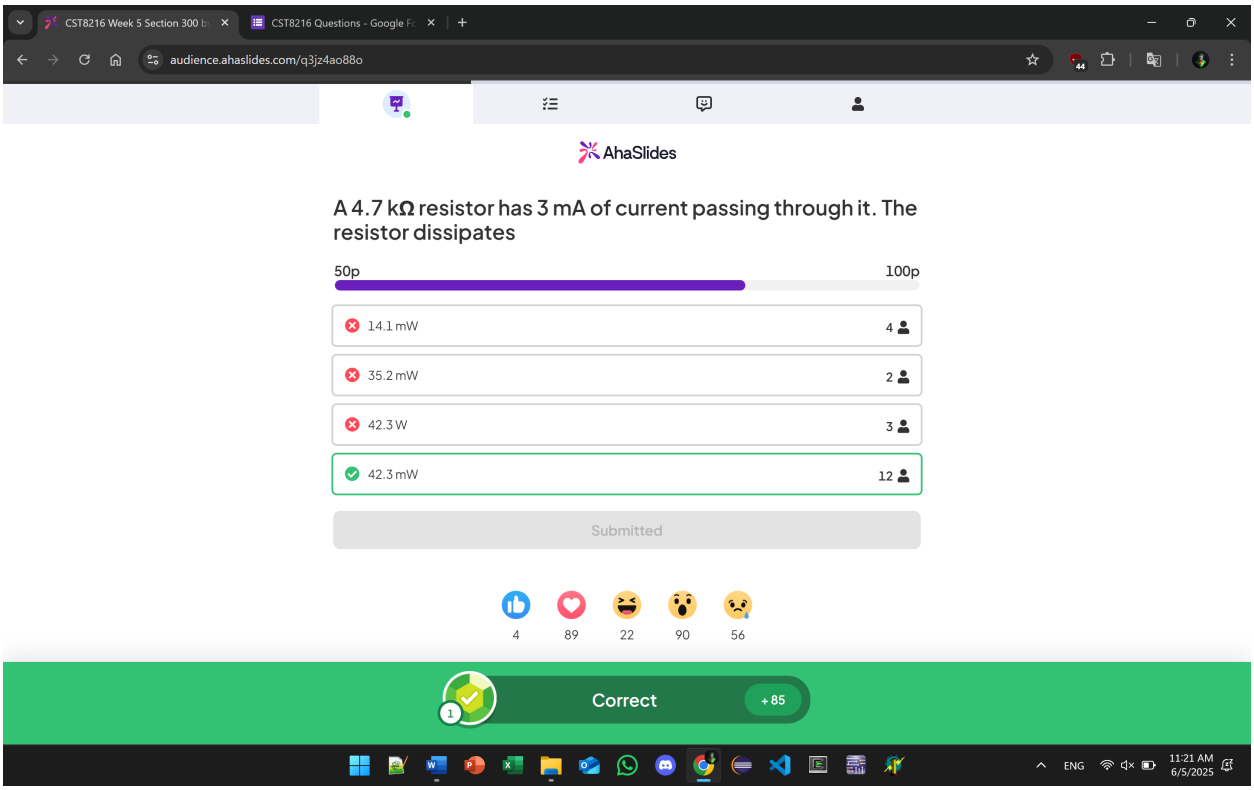


AhaSlides

Convert -65 to 8-bits hexadecimal equivalents



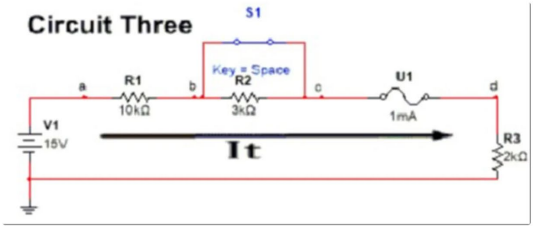

CST8216 Week 5 Section 300 b
audience.ahaslides.com/q3jz4ao88o
AhaSlides
Convert Signed Hexadecimal $E7 to Decimal equivalents
50p
100p
231
4
-25
13
-7
0
25
4
Submitted
4
89
22
89
50
Oh no, streak lost!
+0
ENG
10:59 AM
6/5/2025


CST8216 Week 5 Section 300 b
CST8216 Questions - Google Fo
audience.ahaslides.com/q3jz4ao88o
AhaSlides
A 4.7 kΩ resistor has 3 mA of current passing through it. The resistor dissipates
50p
100p
14.1 mW
4
35.2 mW
2
42.3 W
3
42.3 mW
12
Submitted
4
89
22
90
56
Correct
+85
ENG
11:21 AM
6/5/2025

CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fc

audience.ahaslides.com/q3jz4ao88o

With S1 closed in Circuit Three, the fuse will eventually blow (open). In that condition, it has infinite resistance and we have an OPEN circuit. What is the value of Vc when the fuse is blown?



50p 100p

- 0 V — 4
- 3.462 V — 5
- 2.813 V — 2
- 15 V — 8

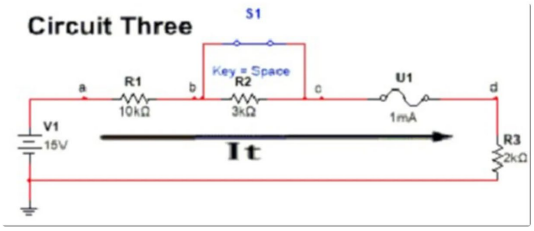
Two in a row! +54

ENG 11:23 AM 6/5/2025

CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fc

audience.ahaslides.com/q3jz4ao88o

With S1 closed in Circuit Three, the fuse will eventually blow (open). In that condition, it has infinite resistance and we have an OPEN circuit. What is the value of It when the fuse is blown?



50p 100p

- 1.145 mA — 2
- 0 A — 13
- 11.54 mA — 1
- 1.154 mA — 1

That's three! +96

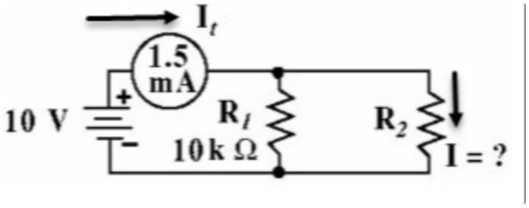
ENG 11:26 AM 6/5/2025

CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fc

audience.ahaslides.com/q3jz4ao88o

AhaSlides

Given the circuit, what is the current flowing through R2?



50p 100p

5.0 mA	0
1.0 mA	2
1.5 mA	6
0.5 mA	11

Fantastic Four! +78

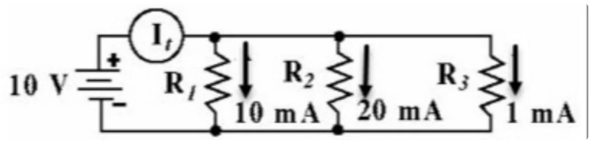
ENG 11:27 AM 6/5/2025

CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fc

audience.ahaslides.com/q3jz4ao88o

AhaSlides

Given the circuit, find the total power.



50p 100p

0.3 W	3
310 mW	13
310 µW	2
0.125 W	0

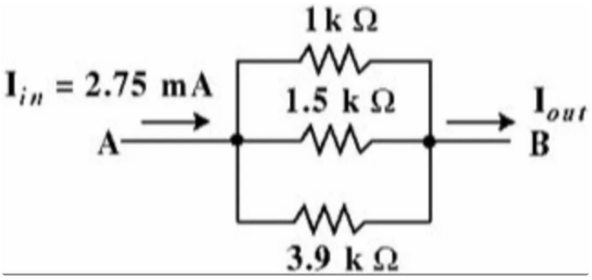
Keep going! +81

ENG 11:30 AM 6/5/2025

CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fc

audience.ahaslides.com/q3jz4ao88o

Given the circuit, what is the output current (Iout)?



50p 100p

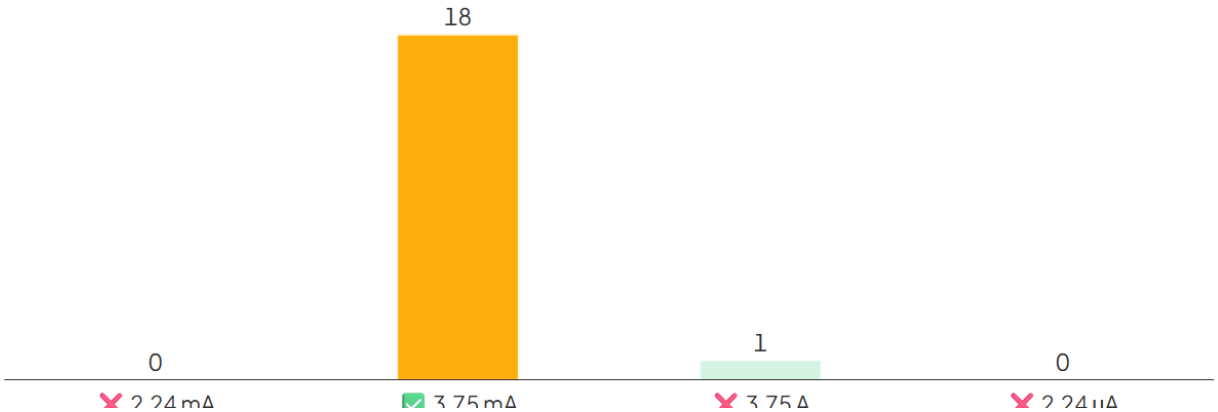
- 2.75 mA — 12
- 0.916 mA — 3
- 1.25 mA — 2
- 0.01 A — 1

6 Amazing! +69

ENG 11:31 AM 6/5/2025

AhaSlides

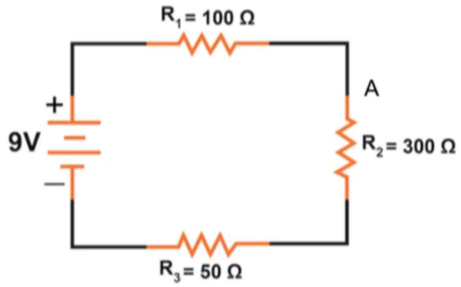
A series circuit consists of three resistors with values of 1.1 kΩ, 4.4 kΩ, and 2.5 kΩ across a 30 V source. The current through the 2.5 kΩ resistor is



CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fc

audience.ahaslides.com/q3jz4ao88o

In the series circuit shown, if R2 is open-circuited, what would the voltage at point A be?



50p 100p

9V	14
0V	4
7V	0
2V	0

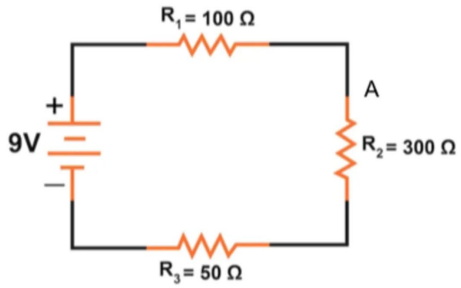
Oh no, streak lost! +0

ENG 11:35 AM 6/5/2025

CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fc

audience.ahaslides.com/q3jz4ao88o

In the series circuit shown, if R2 is short-circuited, what would the voltage at point A be?



50p 100p

9V	8
7V	0
6V	5
3V	4

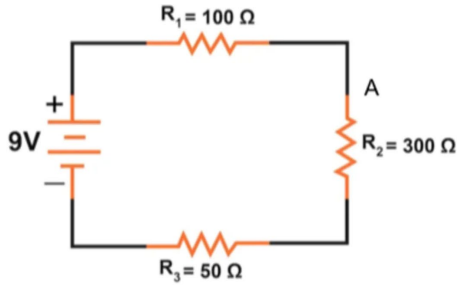
Correct +73

ENG 11:36 AM 6/5/2025

CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fc

audience.ahaslides.com/q3jz4ao88o

In the series circuit shown, if R_1 is open-circuited, what would the voltage at point A be?



50p 100p

✗ 9V	5
✓ 0V	13
✗ 7V	1
✗ 2V	0

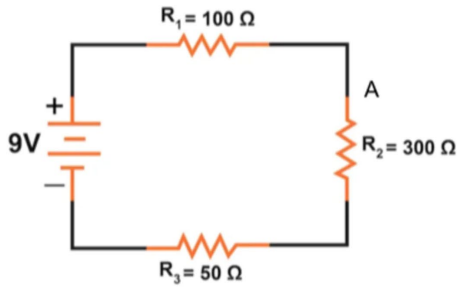
Two in a row! +85

ENG 11:39 AM 6/5/2025

CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fc

audience.ahaslides.com/q3jz4ao88o

In the series circuit shown, if R_1 is short-circuited, what would the voltage at point A be?



50p 100p

✓ 9V	15
✗ 0V	0
✗ 7V	2
✗ 2V	0

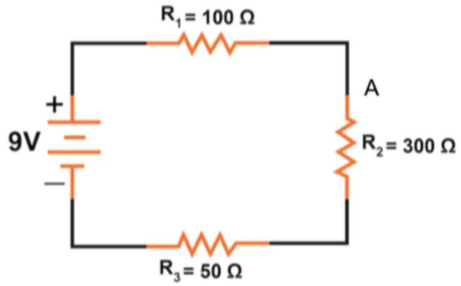
That's three! +63

ENG 11:40 AM 6/5/2025

CST8216 Week 5 Section 300 b | CST8216 Questions - Google Fo

audience.ahaslides.com/q3jz4ao88o

In the series circuit shown, if R3 is short-circuited, what would the voltage at point A be?



50p 100p

7V	1
3V	1
6.75V	11
2.25V	3

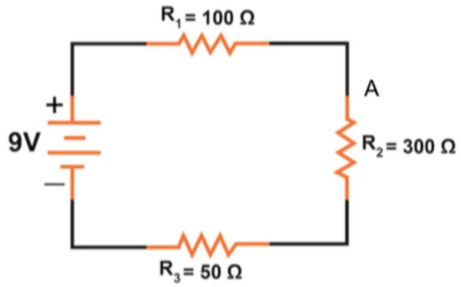
Fantastic Four! +83

ENG 11:42 AM 6/5/2025

CST8216 Week 5 Section 300 b

audience.ahaslides.com/q3jz4ao88o

In the series circuit shown, if R3 is open-circuited, what would the voltage at point A be?



50p 100p

0V	8
9V	6
6.75V	1
7V	1

Oh no, streak lost! +0

ENG 11:43 AM 6/5/2025

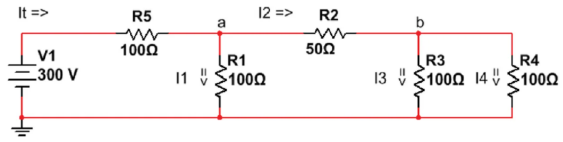
CST8216 Week 5 Section 300 b

audience.ahaslides.com/q3jz4ao88o

AhaSlides

Calculate the value of Rt

Circuit Five



50p 100p

350 Ω	1
150 Ω	15
300 Ω	0
33.3 Ω	1

Submitted

Correct +95

ENG 11:48 AM 6/5/2025